

Process Optimisation & Compliance Analytics

Proposal | Chemical/Process Industry, Kochi, Kerala | 2022 | ■25 Cr

Executive Summary

We propose a targeted process optimisation programme for your Kochi specialty chemicals facility, focused on batch analytics, real-time process parameter monitoring, and regulatory compliance dashboards. Our process industry domain expertise and prior work in the Kerala industrial corridor give us a distinct advantage in delivering fast value.

Client Situation

The facility operates 18 batch reactors producing specialty chemicals for the pharma and agricultural sectors. Batch cycle time variability of $\pm 22\%$ is causing schedule overruns and raw material wastage. Additionally, Kerala Pollution Control Board (KPCB) compliance reporting is manual and error-prone, creating regulatory risk.

South India & Kerala Expertise

Our South India practice, headquartered in Chennai with a delivery presence in Kochi, has delivered three prior engagements in Kerala's process and manufacturing sectors. We understand KPCB compliance requirements, the availability of local engineering talent, and the operational culture of process industries in the Kerala industrial corridor.

Proposed Scope

- Phase 1: Real-time process monitoring — DCS/SCADA data extraction, historian integration
- Phase 2: Batch analytics — cycle-time optimisation models, golden-batch benchmarking
- Phase 3: Compliance dashboard — automated KPCB emission and effluent reporting
- Alarm rationalisation: reduce nuisance alarms by 60%, prioritise actionable events
- Operator advisory system: real-time guidance for batch parameter adjustments
- Integration with ERP (Oracle) for production order and material reconciliation

Commercial Summary

- Total investment: ■25 Cr over 20 months
- Phase 1 (monitoring foundation): ■8 Cr, 6-month delivery
- Phase 2 (batch analytics): ■10 Cr, 8-month delivery
- Phase 3 (compliance): ■7 Cr, 6-month delivery
- Expected ROI: Payback within 26 months through yield improvement and avoided KPCB penalties